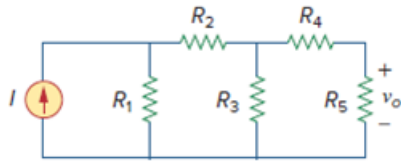


Problem

Using Fig. 4.70, design a problem to help other students better understand linearity.

Figure. 4.70:



Step-by-step solution

Step 1 of 6

Draw the circuit diagram as shown in Figure 1.

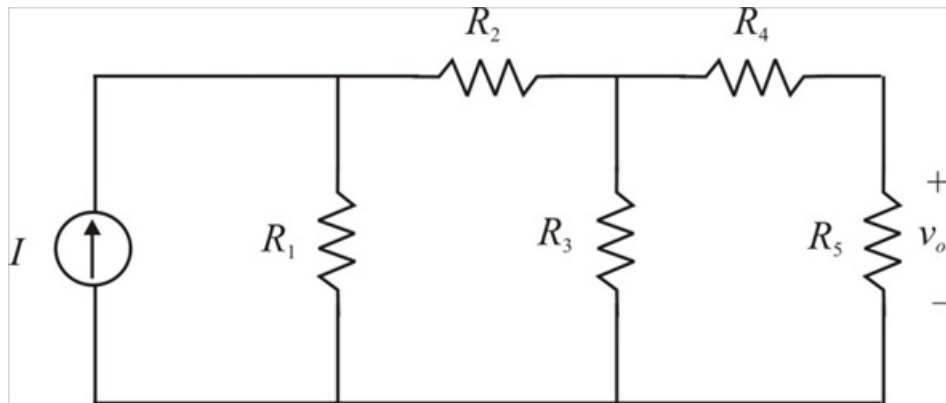


Figure 1

Step 2 of 6

In general, a circuit is a linear if it is both additive and homogeneous.

The additive property requires that the response to a sum of inputs is the sum of the responses to each input applied separately.

The homogeneous property requires that if the input is multiplied by a constant, then the output is multiplied by the same constant.

Let us we find out v_o in the circuit if the source current is reduced to 10 mA.

Let us take some values for the resistances and current source as follows:

$$R_1 = 16\Omega$$

$$R_2 = 10\Omega$$

$$R_3 = 12\Omega$$

$$R_4 = 8\Omega$$

$$R_5 = 4\Omega$$

$$I = 2\text{ A}$$

Step 3 of 6

The circuit is labeled as shown in Figure 2.

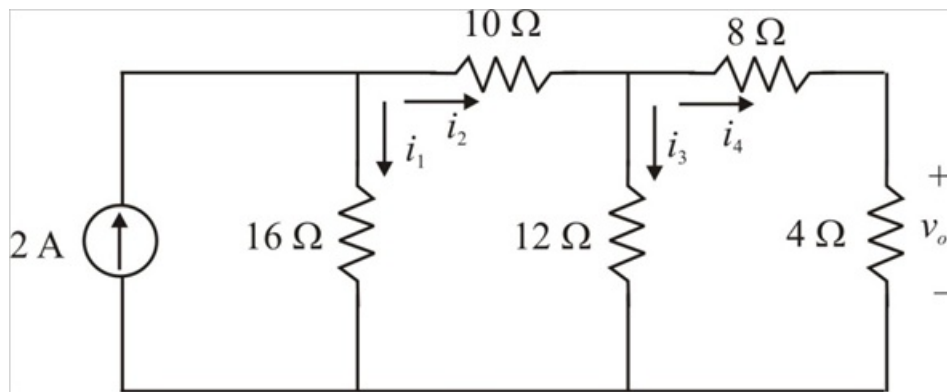


Figure 2

Step 4 of 6

The 4 Ω and 8 Ω resistors are in series. Their combined resistance is:

$$4\Omega + 8\Omega = 12\Omega$$

This 12 Ω and 12 Ω resistors are in parallel. Their combined resistance is:

$$\begin{aligned} 12\Omega \parallel 12\Omega &= \frac{12 \times 12}{12 + 12} \\ &= 6\Omega \end{aligned}$$

This 6 Ω and 10 Ω resistors are in series. Their combined resistance is:

$$6\Omega + 10\Omega = 16\Omega$$

By current division, the current i_1 flowing through 16 Ω resistor is:

$$\begin{aligned} i_1 &= \frac{16}{16+16}(I) \\ &= \frac{16}{16+16}(2) \\ &= 1\text{ A} \end{aligned}$$

Step 5 of 6

By current division, the current i_2 flowing through 10 Ω resistor is:

$$\begin{aligned} i_2 &= \frac{16}{16+16}(I) \\ &= \frac{16}{16+16}(2) \\ &= 1\text{ A} \end{aligned}$$

By current division, the current i_3 flowing through 12 Ω resistor is:

$$\begin{aligned} i_3 &= \frac{12}{12+12}(i_2) \\ &= \frac{12}{12+12}(1) \\ &= 0.5\text{ A} \end{aligned}$$

Step 6 of 6

By current division, the current i_4 flowing through 8 and 4 Ω series resistors is:

$$\begin{aligned} i_4 &= \frac{12}{12+12}(i_2) \\ &= \frac{12}{12+12}(1) \\ &= 0.5 \text{ A} \end{aligned}$$

Therefore the current through 4 Ω resistor is 0.5A

By Ohm's law, the voltage v_o across the 4 Ω resistor can be calculated as:

$$\begin{aligned} v_o &= i_4 R \\ &= (0.5)(4) \\ &= 2 \text{ V} \end{aligned}$$

This shows that assuming source current $I_s = 2 \text{ A}$ gives voltage $v_o = 2 \text{ V}$. The actual source current of 10 mA will give voltage v_o as:

$$\begin{aligned} v_o &= \frac{10 \times 10^{-3}}{2}(2) \\ &= 10 \text{ mV} \end{aligned}$$

Therefore the voltage v_o is 10mV.